

How many labelers do you have?

A closer look at gold-standard labels

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Abstract

The construction of most supervised learning datasets revolves around collecting multiple labels for each instance, then aggregating the labels to form a type of “true” label. We question the wisdom of this pipeline by developing a (stylized) theoretical model of this process and analyzing its statistical consequences, showing how access to non-aggregated label information can make training well-calibrated models more feasible than it is with cleaned labels. The entire story, however, is subtle, and the contrasts between aggregated and fuller label information depend on the particulars of the problem, where estimators that use aggregated information exhibit robust but slower rates of convergence, while estimators that can effectively leverage all labels converge more quickly *if* they have fidelity to (or can learn) the true labeling process. The theory makes several predictions for real-world datasets, including when non-aggregate labels should improve learning performance, which we test to corroborate the validity of our predictions.

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1 Introduction

Challenge datasets drive advances in statistical machine learning [22, 20, 1, 18, 10, 27], making the collection of labeled data essential [11]. While in the past, experts could label reasonably sized datasets, the cost and size of modern datasets often precludes such annotation, motivating crowdsourcing and other dataset generation strategies that aggregate non-expert and expert feedback [9, 33, 27, 8, 12]. Frequently, published data contain aggregated label information; for example, in the dataset [8] of skin lesion images to be classified as malignant, biopsy and histology confirm positive examples, while physician consensus determines negative examples, yet the published data includes no notion of disagreement or uncertainty in these consensus. Similarly, the benchmark CIFAR [18] and ImageNet [10, 27] computer vision challenges include only a single label for each example, in spite of deriving from multiple pieces of feedback per image. Researchers typically perform such aggregation in effort to obtain “high-quality” or “true” labels [10, 27, 9, 33].

Yet most statistical machine learning development does not investigate this full data generating process, assuming only that data comes in the form of (X, Y) pairs of covariates X and labels Y [31, 5, 2, 15]. We argue for a more holistic perspective: that analysis and algorithmic development should focus on the complete statistical learning pipeline, from dataset construction to model output. We question the extent to which such label cleaning strategies are essential, arguing that benchmark datasets ought to include noisy labels as a rule; one may always clean the data in subsequent stages, but sometimes the more nuanced labeling information may yield more accurate downstream models.

We develop a stylized theoretical model to capture uncertainty in the labeling process, allowing us to understand the limitations of and possible improvements to using aggregated data in a statistical learning pipeline. We model each example as a pair $(X_i, (Y_{i1}, \dots, Y_{im}))$

where X_i is a data point and Y_{ij} are noisy binary labels. The basic formulation of our results compares two methods: empirical risk minimization using all labels and empirical risk minimization using majority vote labels Y^+ . While this oversimplifies label aggregation strategies [9, 27, 26], the simplicity (i) allows us to understand fundamental limitations of algorithms based on majority-vote aggregation, (ii) facilitates circumventing these limits via non-aggregated information, and (iii) provides a base for future work: the purposeful simplicity permits classical tools to apply. In these models, our results show that estimators using non-aggregated labels outperform those that use aggregated (majority-vote) labels if the predictive model has fidelity to the data, majority-vote estimators provide robust (but slower) convergence, and these tradeoffs are fundamental. We extend the results to include misspecification, semiparametric scenarios, and simple models of learned annotator reliability.

While our models are stylized, they make testable predictions for real data beyond simplified scenarios, in particular, that models fit on non-aggregated data should both make better-calibrated predictions and have lower classification error than those using cleaned labels. Experiments on two real and one large semisynthetic dataset corroborate the implications of our theory beyond logistic models: majority-vote-based algorithms yield uncalibrated predictors, while algorithms using full label information provide (more) calibrated predictors. The former algorithms exhibit worse classification error in our experiments, with the error gap depending on parameters—e.g., noise—we address in our theoretical models.

1.1 Problem formulation

Consider covariates $X_i \stackrel{\text{iid}}{\sim} \mathbb{P}_X$, $X_i \in \mathbb{R}^d$, in a binary classification problem with m labelers, who annotate points independently through a generalized linear model (GLM) via link functions

$$\sigma_1^*, \dots, \sigma_m^* \in \mathcal{F}_{\text{link}} := \{\sigma : \mathbb{R} \rightarrow [0, 1] \mid \sigma(0) = 1/2, \text{sign}(\sigma(t) - 1/2) = \text{sign}(t)\}.$$

For example, in the logistic model where labelers have identical link, $\sigma_j^*(t) = 1/(1 + e^{-t})$. If $\sigma(t) + \sigma(-t) = 1$ for all $t \in \mathbb{R}$, we say the link function is symmetric, denoting the class of symmetric links by $\mathcal{F}_{\text{link}}^0 \subset \mathcal{F}_{\text{link}}$. Labels then follow the distribution

$$\mathbb{P}_{\sigma, \theta}(Y = y \mid X = x) = \sigma(y\langle \theta, x \rangle) \quad \text{for } y \in \{\pm 1\}, \quad x, \theta \in \mathbb{R}^d. \quad (1)$$

Key to our model, and what allows our analysis, is that we assume labelers use the same linear classifier $\theta^* \in \mathbb{R}^d$, so we obtain conditionally independent labels $Y_{ij} \sim \mathbb{P}_{\sigma_j^*, \theta^*}(\cdot \mid X_i)$. We seek to recover θ^* or the direction $u^* := \theta^* / \|\theta^*\|_2$ from the observations (X_i, Y_{ij}) .

Error measures. We evaluate an estimator $\hat{\theta}$ and associated direction $\hat{u} := \hat{\theta} / \|\hat{\theta}\|_2$ through

$$(E.i) \quad \text{The classification error: } \|u^* - \hat{u}\|_2 = \sqrt{2(1 - \langle u^*, \hat{u} \rangle)}.$$

$$(E.ii) \quad \text{The calibration error: } \|\theta^* - \hat{\theta}\|_2.$$

To motivate these errors, note that for classification, we only need to control the difference between the directions $\hat{u}$ and u^* , while calibration—that for a data point X , the value $\sigma_j^*(\langle \hat{\theta}, X \rangle) \approx \sigma_j^*(\langle \theta^*, X \rangle)$ —requires controlling the error in $\hat{\theta}$ as an estimate of θ^* . If X is rotationally symmetric, then $\mathbb{P}(\text{sign}(\langle X, u \rangle) \neq \text{sign}(\langle X, u^* \rangle)) = \frac{1}{\pi} \cos^{-1} \langle u, u^* \rangle$ for any unit vectors u, u^* ; because $\cos^{-1} t = (1 + o(1))\sqrt{2(1 - t)}$ as $t \uparrow 1$, the classification error (E.i), $\|u^* - u\|_2 = \sqrt{2(1 - \langle u, u^* \rangle)}$, is asymptotically equivalent to the angle between u and u^* .

Estimators. We consider two types of estimators: one using processed labels Y_i^+ for each example X_i , and the other using each label $Y_{i1}, \dots, Y_{im}$. To center discussion, we instantiate this temporarily via logistic regression for the logistic link $\sigma^{\text{lr}}(t) = \frac{1}{1 + e^{-t}}$, defining

$$\ell_{\theta}^{\text{lr}}(y \mid x) = -\log \mathbb{P}_{\sigma^{\text{lr}}, \theta}(y \mid x) = \log(1 + e^{-y\langle x, \theta \rangle}).$$

In the non-aggregated model, we let $\mathbb{P}_{n,m}$ be the empirical measure on $\{(X_i, (Y_{i1}, \dots, Y_{im}))\}$ and consider the logistic regression estimator (i.e., the MLE assuming the logistic model)

$$\hat{\theta}_{n,m}^{\text{lr}} = \underset{\theta}{\operatorname{argmin}} \left\{ \mathbb{P}_{n,m} \ell_{\theta}^{\text{lr}} = \frac{1}{nm} \sum_{i=1}^n \sum_{j=1}^m \ell_{\theta}^{\text{lr}}(Y_{ij} \mid X_i) \right\}. \quad (2)$$

We focus on the simplest aggregation strategy, where example i has majority vote label

$$Y_i^+ = \text{maj}(Y_{i1}, \dots, Y_{im}) := \text{sign} \left(\sum_{j=1}^m Y_{ij} \right) \quad \text{where} \quad \text{sign}(0) \sim \text{Ber}(1/2)$$

Then letting $\mathbb{P}_{n,m}^+$ be the empirical measure on $\{(X_i, Y_i^+)\}$, the majority-vote estimator solves

$$\hat{\theta}_{n,m}^{\text{mv}} = \underset{\theta}{\operatorname{argmin}} \left\{ \mathbb{P}_{n,m}^+ \ell_{\theta}^{\text{lr}} = \frac{1}{n} \sum_{i=1}^n \ell_{\theta}^{\text{lr}}(Y_i^+ \mid X_i) \right\}. \quad (3)$$

Method (3) acts as our proxy for the “standard” data analysis pipeline, with cleaned labels, while method (2) is our proxy for non-aggregated methods using all labels. More general formulations than the majority vote (3) could allow complex aggregation strategies, e.g., crowdsourcing, but we simplify to capture what we view as the essentiae for problems (e.g. CIFAR [18] or ImageNet [27]) where only aggregated label information is available.

Our main results characterize the asymptotics of the estimators $\hat{\theta}_{n,m}^{\text{mv}}$ and $\hat{\theta}_{n,m}^{\text{lr}}$. Under appropriate assumptions on the data generating mechanisms (1), which include misspecification, we (i) provide consistency results that elucidate the limits for $\hat{\theta}_{n,m}^{\text{mv}}$, $\hat{\theta}_{n,m}^{\text{lr}}$, and a few more general estimators, and (ii) evaluate their limit distributions via asymptotic normality. The latter allows comparison of the estimators through their limiting covariances, which exhibit (to us) interestingly varying dependence on m and the scaling of the true parameter θ^* .

1.2 Summary of results and implications

Asymptotic characterization of the estimators (2)–(3) provides our main avenue to highlighting the import of multiple labels. Table 1 summarizes our results, comparing the performance

	MLE-type estimator (2)	majority-vote estimator (3)
single link	Classification: $r(m) \asymp \frac{1}{m}$	Classification: $r(m) \asymp \frac{1}{\sqrt{m}}$
(well-specified)	Calibration: $t(m) = 1$	Calibration: $t(m) \asymp \sqrt{m}$
mis-specified or	Calibration: $t(m) \asymp 1$	Calibration: $t(m) \asymp \sqrt{m}$
multiple links	Classification: $r(m) \asymp 1$	Classification: $r(m) \asymp 1/\sqrt{m}$

Table 1. Comparison of multi-label and majority-vote estimators for $X \sim \mathcal{N}(0, I_d)$ according to the scalings in classification error $r(m)$ and calibration error $t(m)$ that the limits (4) identify. If $t(m) \gg 1$, the model is uncalibrated; if the rate $r(m)$ is small, the model achieves lower asymptotic variance. Boldface indicates the associated estimator is (rate) optimal.

of the MLE-type (2) and majority vote (3) estimators for the classification (E.i) and calibration (E.ii) errors. We highlight estimator performance in two scenarios: “single link” corresponds to the case that the loss is well-specified, while “mis-specified” corresponds to the case that $\ell_\theta(y | x) \neq -\log \mathbb{P}_{\sigma, \theta}(y | x)$, which may occur when labelers have distinct link functions σ or the model is incorrect. In all cases, the estimate $\hat{\theta}$ and vector $\hat{u} = \hat{\theta} / \|\hat{\theta}\|_2$ are asymptotically normal around a scalar multiple of θ^* , with differing scaling and asymptotic variance. With m labelers, each estimator yields a rate $r(m)$ and scale $t(m)$ for which

$$\sqrt{n}(\hat{u} - u^*) \rightsquigarrow \mathcal{N}(0, r(m)\Sigma_n) \quad \text{and} \quad \sqrt{n}(\hat{\theta} - t(m)\theta^*) \rightsquigarrow \mathcal{N}(0, \Sigma_u), \quad (4)$$

where Σ_n and Σ_u are “normalized” and “unnormalized” covariances, respectively. The classification error (E.i) is small whenever $r(m) \ll 1$, while the calibration error (E.ii) is small when $t(m) \approx 1$, and Table 1 shows the scalings (4) for the different estimators.

Improved performance using multiple labels for well-specified models We begin in Section 2 by specializing our main results to the logistic model. We show that the MLE (2) is calibrated and enjoys faster rates of convergence (in m) than the majority-vote estimator (3). The improvements depend on the underlying distribution, and Propositions 1 and 2 connect

them to Mammen-Tsybakov-type noise conditions. In “standard” cases, the improvement scales as $\sqrt{m}$; for problems with little classification noise, the majority vote estimator becomes brittle (relatively slower), while the convergence rate gap decreases for noisier problems.

Robustness of majority-vote estimators Our results also evidence support for majority vote estimators (3). Section 3 provides master results holding for well- and mis-specified losses, which highlight the robustness of the majority vote estimator (cf. Table 1). The faster rates MLE-type estimators (2) enjoy when the model is correct break down in ways we make precise for mis-specified models; in contrast, majority-vote-type estimators (3) maintain their (not quite optimal) improved convergence guarantees, even under mis-specification, suggesting practical value to cleaning data when correctly modeling uncertainty is impossible.

Fundamental limits of majority vote Via local asymptotic minimax theory, Section 4 develops fundamental limits for estimation given only majority vote labels (X, Y^+) , which coincide with our results in Section 2. These results highlight how cleaned labels force slower convergence in the number of labelers m while reinforcing the claims of robustness. In particular, while a mis-specified link function introduces structural bias that MLE-type estimators using all labels cannot overcome, majority vote-based estimators achieve rate-optimal convergence for procedures receiving cleaned labels Y^+ (making the rate $r(m)$ in the bottom right of Table 1 sharp).

Experimental evaluation Our theoretical results predict two outcomes: (i) *if* the model has fidelity to the labeling process, then using all noisy labels should yield better estimates than procedures using cleaned labels, and (ii), that majority vote estimators should be more robust. In Section 5, we thus provide real and semi-synthetic experiments that corroborate

the theoretical predictions; the experiments also highlight a need, which some researchers are now beginning to address [12], for more applied dataset development so that we can evaluate the consequences of filtering and otherwise manipulating the data we use.

Semi-parametric approaches In our discussion (Section 6), we include an exemplar approach for estimation via semi-parametric approaches, where one estimates labelers’ links and reliability. While not our main focus, the results provide potential insights into estimation using existing crowdsourcing techniques, where black box algorithms provide measures of labeler reliability that one may incorporate to achieve rate-optimal estimation.

1.3 Related work

To make our theoretical model tractable, we capture only a few of the complexities inherent in dataset construction. Label aggregation strategies often attempt to evaluate labeler uncertainty, dating to Dawid and Skene [9], who study labeler uncertainty estimation to overcome noise in clinical patient measurements. With the rise of crowdsourcing, such reliability models have attracted interest, with approaches for optimal budget allocation [17], addressing untrustworthy or malicious labelers [7], and more broadly a line of applied [10, 27] and theoretical work studying crowd labeling and aggregation [33, 32, 26].

Many of these focus, however, on obtaining a single trustworthy label for each example. Our work takes a different perspective. First, we target statistical analysis for the full learning pipeline—understanding the landscape of the learning problem with multiple labels for each example—as opposed to obtaining only clean labels. Moreover, we argue for calibrating the resulting predictive model, which cleaned labels necessarily impede. We therefore adopt a perspective similar to the applied works [23, 24], which highlight how incorporating human

uncertainty into learning pipelines can make classification “more robust” [23]. Instead of splitting the learning procedure into two phases, where the first aggregates labels and the second trains, we use non-aggregated labels throughout learning. Platanios et al. [24] directly model labeler reliability via a richer model than our stylized scenarios, but our approach allows theoretically precise description of the methods’ limiting behavior.

Our results complement a growing body of work in machine learning. The Neural Information Processing Systems conference includes a Datasets and Benchmarks track, as they “are crucial for the development of machine learning methods” [3], building off the long history of dataset-driven development in machine learning [14]. We attempt to lay statistical foundations for this line of work.

Notation For a matrix M , $\|M\|$ denotes its spectral norm and $M^\dagger$ its Moore-Penrose pseudo inverse. For a unit vector u , $\mathbf{P}_u^\perp := I - uu^\top$ projects to the space orthogonal to $\text{span}\{u\}$. We use the empirical process notation $\mathbb{P}Z = \int z d\mathbb{P}(z)$ and $\rightsquigarrow$ denotes convergence in distribution. We write $f(x) \asymp g(x)$ for $x \in \mathbb{R}_+$ if there exist numerical constants $0 < c_1, c_2 < \infty$ and $x_0 \geq 0$ such that $c_1|g(x)| \leq |f(x)| \leq c_2|g(x)|$ for $x \geq x_0$, and $c = o_m(1)$ means $c \rightarrow 0$ as $m \rightarrow \infty$.

2 The well-specified logistic model

We begin in a setting that allows the cleanest and most precise comparisons between a method using aggregated labels (3) and one without (2) by considering the logistic link for (1),

$$\sigma^{\text{lr}}(t) = \frac{1}{1 + e^{-t}} \in \mathcal{F}_{\text{link}}^0.$$

In particular, we assume identical links $\sigma_1^\star = \dots = \sigma_m^\star = \sigma^{\text{lr}}$ and an i.i.d. sample $X_1, \dots, X_n$, where for each $i \in [n]$ we draw $(Y_{i1}, \dots, Y_{im}) \stackrel{\text{iid}}{\sim} \mathbb{P}_{\sigma^{\text{lr}}, \theta^\star}(\cdot \mid X_i)$ for a true vector $\theta^\star$.

2.1 The isotropic Gaussian case

To deliver the general taste of our results on the performance of the full information (2) and majority vote (3) approaches, we start by studying the simplest case when $X \sim \mathbf{N}(0, I_d)$.

Performance with non-aggregated data. The non-aggregated MLE $\hat{\theta}_{n,m}^{\text{lr}}$ in Eq. (2) admits a standard analysis, which we state as a corollary of Proposition 1 to come.

Corollary 1. *Let $X \sim \mathbf{N}(0, I_d)$ and $t^* = \|\theta^*\|_2$. The maximum likelihood estimator $\hat{\theta}_{n,m}^{\text{lr}}$ is consistent, with $\hat{\theta}_{n,m}^{\text{lr}} \xrightarrow{p} \theta^*$, and for $\mathbf{P}_{u^*}^\perp = I_d - u^* u^{*\top}$,*

$$\sqrt{n}(\hat{u}_{n,m}^{\text{lr}} - u^*) \rightsquigarrow \mathbf{N}(0, m^{-1} \cdot (t^*)^{-2} \cdot \mathbb{E}[\sigma^{\text{lr}}(\langle \theta^*, X \rangle)(1 - \sigma^{\text{lr}}(\langle \theta^*, X \rangle))]^{-1} \mathbf{P}_{u^*}^\perp).$$

So the non-aggregated MLE is calibrated, recovering θ^* , and enjoys convergence rates scaling as $1/\sqrt{nm}$, so that increasing the number of labels m linearly improves convergence rate.

Performance with majority-vote aggregation. The analysis of the majority vote estimator (3) requires more care, though the assumption that $X \sim \mathbf{N}(0, I_d)$ allows us to calculate the limits explicitly. In brief, we show that when X is Gaussian, the estimator is mis-calibrated and its classification error converges more slowly in m than the non-aggregated classifier. The basic idea is that the majority vote classifier should still point in the direction of θ^* , but it should be “calibrated” to the probability of a majority of m labels being correct.

We follow this idea and sketch the derivation here, as it is central to all of our theorems, then state a companion corollary. Each result depends on the probability

$$\mathbb{P}(Y^+ = \text{sign}(\langle x, \theta^* \rangle) \mid x) = \rho_m(|\langle x, \theta^* \rangle|)$$

of obtaining a correct label using majority vote, where ρ_m is the binomial probability

$$\rho_m(t) = \mathbb{P}\left(\text{Bin}\left(m, \frac{1}{1 + e^{-|t|}}\right) \geq \frac{m}{2}\right) = \sum_{i=\lceil m/2 \rceil}^m \binom{m}{i} \left(\frac{1}{1 + e^{-|t|}}\right)^i \left(\frac{e^{-|t|}}{1 + e^{-|t|}}\right)^{m-i}, \quad (5)$$

when m is odd. (When m is even the sum has the asymptotically negligible additional additive term $\frac{1}{2} \binom{m}{m/2} \frac{e^{-m|t|/2}}{(1+e^{-|t|})^m}$.) Key to the coming result is a parameter that approximately equalizes binomial (majority vote) and logistic (Bernoulli) probabilities, so we define

$$h_m(t) = \mathbb{E} [|Z|(1 - \rho_m(t^*|Z|))] - \mathbb{E} \left[\frac{|Z|}{1 + e^{t|Z|}} \right], \quad \text{for } Z \sim \mathbf{N}(0, 1). \quad (6)$$

We use h to find the minimizer of population loss $L_m^{\text{mv}}(\theta) = \mathbb{E}[\ell_\theta^{\text{r}}(Y^+ | X)]$ via the ansatz $\theta = tu^*$ for some $t > 0$. Using the definition (5) of ρ_m , for $S_\star = \text{sign}(\langle X, \theta^* \rangle)$,

$$L_m^{\text{mv}}(\theta) = \mathbb{E} [\log(1 + e^{-S_\star \langle X, \theta \rangle}) \cdot \rho_m(t^*|\langle X, u^* \rangle|)] + \mathbb{E} [\log(1 + e^{S_\star \langle X, \theta \rangle}) \cdot (1 - \rho_m(t^*|\langle X, u^* \rangle|))],$$

where we recall $t^* = \|\theta^*\|_2$, and compute the gradient for $\theta = tu^*$ via

$$\nabla L_m^{\text{mv}}(\theta) = -\mathbb{E} \left[\frac{S_\star \rho_m(t^*|\langle X, u^* \rangle|)}{1 + \exp(S_\star \langle X, \theta \rangle)} X \right] + \mathbb{E} \left[\frac{S_\star \exp(S_\star \langle X, \theta \rangle) (1 - \rho_m(t^*|\langle X, u^* \rangle|))}{1 + \exp(S_\star \langle X, \theta \rangle)} X \right].$$

Setting $Z = \langle X, u^* \rangle$ to decompose X into the independent sum $X = (X - u^*Z) + u^*Z$,

$$\begin{aligned} \nabla L_m^{\text{mv}}(tu^*) &\stackrel{(i)}{=} -\mathbb{E} \left[\frac{\text{sign}(Z)}{1 + \exp(t|Z|)} \rho_m(t^*|Z|) X \right] + \mathbb{E} \left[\frac{\text{sign}(Z) \exp(t|Z|)}{1 + \exp(t|Z|)} (1 - \rho_m(t^*|Z|)) X \right] \\ &= \left(\mathbb{E} [|Z|(1 - \rho_m(t^*|Z|))] - \mathbb{E} \left[\frac{|Z|}{1 + e^{t|Z|}} \right] \right) u^* = h_m(t) u^*, \end{aligned} \quad (7)$$

where in (i) we substitute $S_\star = \text{sign}(Z)$. As we present in Corollary 2, $h_m(t) = 0$ has a unique solution $t_m \asymp \sqrt{m}$, so $t_m u^*$ necessarily uniquely minimizes the population loss L_m^{mv} .

By completing the calculations for the value of t_m above and leveraging asymptotic normality, we have the following special case of Proposition 2 to come.

Corollary 2. *Let $X \sim \mathbf{N}(0, I_d)$ and $t^* = \|\theta^*\|_2$. There are numerical constants $a, b > 0$ such that for the function $h = h_m$ in (6), there is a unique $t_m \geq t_1 = t^*$ solving $h(t_m) = 0$ and*

$$\widehat{\theta}_{n,m}^{\text{mv}} \xrightarrow{p} t_m u^* \quad \text{and} \quad \lim_{m \rightarrow \infty} \frac{t_m}{t^* \sqrt{m}} = a.$$

Moreover, there exists $C_m(t) = \frac{b}{t\sqrt{m}}(1 + o_m(1))$ such that $\widehat{u}_{n,m}^{\text{mv}} = \widehat{\theta}_{n,m}^{\text{mv}} / \|\widehat{\theta}_{n,m}^{\text{mv}}\|_2$ satisfies

$$\sqrt{n} (\widehat{u}_{n,m}^{\text{mv}} - u^*) \rightsquigarrow \mathbf{N}(0, C_m(t^*) \mathbf{P}_{u^*}^\perp).$$

It is instructive to compare the rates of this estimator to the rates for the non-aggregated MLE in Corollary 1. First, the non-aggregated estimator is calibrated in that $\hat{\theta}_{n,m}^{\text{lr}} \rightarrow \theta^*$, in contrast to the majority-vote estimator, which roughly “calibrates” to the probability majority vote is correct (cf. (7)) via the convergence $\hat{\theta}_{n,m}^{\text{mv}} \rightarrow c\sqrt{m}\theta^*$ as $n \rightarrow \infty$. The scaling of C_m in Corollary 2 also shows that the majority-vote estimator exhibits worse convergence rates by a factor of $\sqrt{m}$ than the estimator $\hat{\theta}_{n,m}^{\text{lr}}$: regardless of the dimension d , for constants c^{lr} and c^{mv} that depend only on $t^* = \|\theta^*\|_2$, the asymptotic variances differ by $\sqrt{m}$, as

$$\sqrt{n}(\hat{u}_{n,m}^{\text{lr}} - u^*) \rightsquigarrow \mathbf{N}(0, m^{-1}c^{\text{lr}}\mathbf{P}_{u^*}^\perp) \quad \text{while} \quad \sqrt{n}(\hat{u}_{n,m}^{\text{mv}} - u^*) \rightsquigarrow \mathbf{N}(0, m^{-1/2}c^{\text{mv}}\mathbf{P}_{u^*}^\perp).$$

To obtain intuition for this result via comparison with Corollary 1, consider the Fisher information $\mathbb{E}[\sigma^{\text{lr}}(\langle \theta^*, X \rangle)(1 - \sigma^{\text{lr}}(\langle \theta^*, X \rangle))XX^\top]$ for logistic regression. Letting $Z^* = \langle \theta^*, X \rangle$ be the predicted margin, there is only “information” available for an estimator when $\sigma^{\text{lr}}(Z^*)$ is near $\frac{1}{2}$, as otherwise $\sigma^{\text{lr}}(Z^*)(1 - \sigma^{\text{lr}}(Z^*)) \approx 0$; under majority vote, we analogously require $\mathbb{P}(\text{Bin}(m, \sigma^{\text{lr}}(Z^*)) > \frac{m}{2}) \approx \frac{1}{2}$, which in turn means that we need $\sigma^{\text{lr}}(Z^*) \in \frac{1}{2} \pm O(\frac{1}{\sqrt{m}})$ by standard binomial concentration, or $Z^* = \langle \theta^*, X \rangle \in \pm O(\frac{1}{\sqrt{m}})$. This occurs on about $1/\sqrt{m}$ fraction of the sample, so of $n \cdot m$ total observations, majority vote “loses” a factor $\sqrt{m}$.

2.2 Comparisons for more general feature distributions

The key to the preceding results is that X decomposes into components aligned with θ^* and independent noise. By abstracting away the Gaussianity assumptions, we may carefully track margins and label noise, and their roles in the merits of maximum-likelihood-type (full label information) estimators relative to those using majority-vote labels. The results follow from the master theorems in the sequel, which rely on the following independence

Assumption A1. *The covariates X have non-singular covariance Σ and decompose as a*

sum of independent random vectors in the span of u^* and its complement

$$X = W + Zu^*, \quad \text{where } W \perp Z, \quad \langle W, u^* \rangle = 0, \quad \mathbb{E}[W] = 0, \quad \mathbb{E}[Z] = 0.$$

We adopt [Mammen and Tsybakov](#)'s measurement [\[21\]](#) of classification difficulty through the gap $\mathbb{P}(Y = 1 \mid X = x) - 1/2$: for a *noise exponent* $\beta \in (0, \infty)$, we consider the condition

$$\mathbb{P}(|\mathbb{P}(Y = 1 \mid X) - 1/2| \leq \epsilon) = O(\epsilon^\beta). \quad (\mathbf{M}_\beta)$$

As $\beta \uparrow \infty$ examples are less likely to have small margin, and the limit $\beta = \infty$ gives the hard margin $|\mathbb{P}(Y = 1 \mid X) - \frac{1}{2}| \geq \epsilon$ for small ϵ . Under the independent decomposition Assumption [A1](#), the noise condition $(\mathbf{M}_\beta)$ solely depends on the covariate's projection onto the signal Z , motivating the following assumption on Z .

Assumption A2. *For a given $\beta > 0$, the signal variable Z is (β, c_Z) -regular, meaning that the absolute value $|Z|$ has density $p(z)$ on $(0, \infty)$, no point mass at 0, and satisfies*

$$\sup_{z \in (0, \infty)} z^{1-\beta} p(z) < \infty, \quad \lim_{z \rightarrow 0} z^{1-\beta} p(z) = c_Z \in (0, \infty).$$

Concretely, when the features X are isotropic Gaussian, $Z \sim \mathbf{N}(0, 1)$ and $\beta = 1$.

For a link σ differentiable at 0 with $\sigma'(0) > 0$, we have $\mathbb{P}_\sigma(Y = 1 \mid Z = z) = \sigma(t^*z) = \frac{1}{2} + \sigma'(0)t^*z + o(t^*z)$, so the Mammen-Tsybakov noise condition $(\mathbf{M}_\beta)$ is equivalent to

$$\mathbb{P}(|t^*Z| \leq \epsilon) = O(\epsilon^\beta).$$

Thus, under Assumption [A2](#), condition $(\mathbf{M}_\beta)$ holds, as by dominated convergence

$$\mathbb{P}(|t^*Z| \leq \epsilon) = \int_0^{\epsilon/t^*} p(z) dz = \int_0^{\epsilon/t^*} c_Z(1 + o_\epsilon(1)) z^{\beta-1} dz = \frac{c_Z}{\beta} \epsilon^\beta \cdot (1 + o_\epsilon(1)).$$

We provide extensions of Corollaries [1](#) and [2](#) in the more general cases the noise exponent β allows. The maximum likelihood estimator retains its convergence guarantees in this setting, and we can generalize the final claim of Corollary [1](#) (see Appendix [F.1](#) for a proof):

Proposition 1. *Let Assumptions A1 and A2 hold for some $\beta > 0$ and $t^\star = \|\theta^\star\|_2$. Let $L^{\text{lr}}(\theta) = \mathbb{E}[\ell_\theta^{\text{lr}}(Y | X)]$ be the population loss. Then maximum likelihood estimator (2) satisfies*

$$\sqrt{n} \left(\hat{\theta}_{n,m}^{\text{lr}} - \theta^\star \right) \rightsquigarrow \mathbf{N}(0, m^{-1} \nabla^2 L^{\text{lr}}(\theta^\star)^{-1}).$$

Moreover,

$$\sqrt{n} \left(\hat{u}_{n,m}^{\text{lr}} - u^\star \right) \rightsquigarrow \mathbf{N} \left(0, m^{-1} \cdot (t^\star)^{-2} \mathbf{P}_{u^\star}^\perp \nabla^2 L^{\text{lr}}(\theta^\star)^{-1} \mathbf{P}_{u^\star}^\perp \right),$$

and there exist $C(t)$ and constant $b > 0$ such that $C(t) = \frac{b}{t^{2-\beta}}(1 + o_t(1))$ and

$$\sqrt{n} \left(\hat{u}_{n,m}^{\text{lr}} - u^\star \right) \rightsquigarrow \mathbf{N} \left(0, m^{-1} \cdot C(t^\star) \left(\mathbf{P}_{u^\star}^\perp \Sigma \mathbf{P}_{u^\star}^\perp \right)^\dagger \right).$$

For majority-vote aggregation, we can generalize Corollary 2 to have $t_m \asymp \sqrt{m}$, but now the noise exponent β determines the convergence rate (see Appendix F.2 for a proof):

Proposition 2. *Let Assumptions A1 and A2 hold for some $\beta \in (0, \infty)$ and $t^\star = \|\theta^\star\|_2$. Let $h = h_m$ be the function (6) with signal covariate Z defined in Assumption A1. Then there are constants $a, b > 0$, depending only on β and c_Z , such that the following hold: there is a unique $t_m \geq t_1 = t^\star$ solving $h(t_m) = 0$ and for this t_m we have both*

$$\hat{\theta}_{n,m}^{\text{mv}} \xrightarrow{p} t_m u^\star \quad \text{and} \quad \lim_{m \rightarrow \infty} \frac{t_m}{t^\star \sqrt{m}} = a.$$

Moreover, there exists a function $C_m(t) = \frac{b}{(t\sqrt{m})^{2-\beta}}(1 + o_m(1))$ as $m \rightarrow \infty$ such that

$$\sqrt{n} \left(\hat{u}_{n,m}^{\text{mv}} - u^\star \right) \rightsquigarrow \mathbf{N} \left(0, C_m(t^\star) \left(\mathbf{P}_{u^\star} \Sigma \mathbf{P}_{u^\star} \right)^\dagger \right).$$

Paralleling the discussion in Section 2.1, we may compare the performance of the MLE $\hat{\theta}_{n,m}^{\text{lr}}$ and the majority-vote estimator $\hat{\theta}_{n,m}^{\text{mv}}$. When the classification problem is hard—meaning that β in Condition (M $_\beta$) is near 0—the aggregation in the majority vote estimator still allows convergence (nearly) as quickly as the non-aggregated estimator; the problem is so noisy that data “cleaning” by aggregation is helpful. For easier problems, where $\beta \gg 0$, the

gap between them grows; this is sensible, as aggregation is likely to force a dataset to be separable, thus making fitting methods unstable (a minimizer may even fail to exist).

3 Label aggregation and misspecified model

The logistic link provides clean interpretation and results, but we can move beyond it to more realistic cases where labelers use distinct links; to allow precise statements, we still assume the same linear term $x \mapsto \langle \theta^*, x \rangle$ for each labeler's generalized linear model. We study generalizations of the maximum likelihood and majority vote estimators (2) and (3), highlighting dependence on link fidelity. In this setting, there are m (unknown and possibly distinct) link functions σ_i^* , $i = 1, 2, \dots, m$. We show that the majority-vote estimator $\hat{\theta}_{n,m}^{\text{mv}}$ enjoys better robustness to model mis-specification than the non-aggregated estimator $\hat{\theta}_{n,m}^{\text{lr}}$, though both use identical losses. In particular, our main result in this section implies

$$\sqrt{n}(\hat{u}_{n,m}^{\text{lr}} - u^*) \rightsquigarrow \mathbf{N}(0, c^{\text{lr}}\Sigma) \quad \text{while} \quad \sqrt{n}(\hat{u}_{n,m}^{\text{mv}} - u^*) \rightsquigarrow \mathbf{N}(0, m^{-1/2}c^{\text{mv}}\Sigma),$$

where c^{lr} and c^{mv} are problem-dependent constants. In contrast to the previous section, the majority-vote estimator enjoys roughly $\sqrt{m}$ -faster rates than the non-aggregated estimator, maintaining its (slow) improvement with m , which the MLE loses to misspecification.

To set the stage for our results, we define the general link-based loss

$$\ell_{\sigma,\theta}(y \mid x) := - \int_0^{y\langle\theta,x\rangle} \sigma(-v)dv.$$

We then consider the general multi-label and majority-vote estimators using the loss $\ell_{\sigma,\theta}$,

$$\hat{\theta}_{n,m}(\sigma) := \underset{\theta}{\operatorname{argmin}} \mathbb{P}_{n,m} \ell_{\sigma,\theta}, \quad \hat{\theta}_{n,m}^{\text{mv}}(\sigma) := \underset{\theta}{\operatorname{argmin}} \bar{\mathbb{P}}_{n,m} \ell_{\sigma,\theta}. \quad (8)$$

For the logistic link $\sigma = \sigma^{\text{lr}}$, we recover the results for the logistic loss $\ell_{\sigma^{\text{lr}},\theta}(y \mid x) = \ell_{\theta}^{\text{lr}}(y \mid x)$ in Section 2. We suppress the link dependence to write $\hat{\theta}_{n,m}, \hat{\theta}_{n,m}^{\text{mv}}$ when the context is clear.

3.1 Master results

We provide master results as a foundation for our convergence rate analyses, and by a bit of notational chicanery, we consider the majority vote and multiple (non-aggregated) simultaneously. When the estimator uses the majority vote Y^+ , let

$$\varphi_m(t) = \rho_m(t)\mathbb{1}\{t \geq 0\} + (1 - \rho_m(t))\mathbb{1}\{t < 0\}, \text{ where } \rho_m(t) := \mathbb{P}(Y^+ = \text{sign}(t) \mid \langle X, \theta^* \rangle = t),$$

and in the case that the estimator uses each label from the m labelers, let

$$\varphi_m(t) = \frac{1}{m} \sum_{j=1}^m \sigma_j^*(t) = \frac{1}{m} \sum_{j=1}^m \mathbb{P}(Y_j = 1 \mid \langle X, \theta^* \rangle = t).$$

In either case, we then see that the population loss with the link-based loss $\ell_{\sigma, \theta}$ becomes

$$L(\theta, \sigma) = \mathbb{E}[\ell_{\sigma, \theta}(1 \mid X)\varphi_m(\langle X, \theta^* \rangle) + \ell_{\sigma, \theta}(-1 \mid X)(1 - \varphi_m(\langle X, \theta^* \rangle))], \quad (9)$$

where we have taken a conditional expectation given X . We assume Assumption A1 holds, that X decomposes into the independent sum $X = Zu^* + W$ with $W \perp u^*$, and the true link functions $\sigma_j^* \in \mathcal{F}_{\text{link}}$. We further impose the following assumption for the model link, which guarantees a non-vanishing gap to the margin when $|Z|$ is large.

Assumption A3. *For $s \in \{-1, 1\}$, the model link function σ satisfies $\lim_{t \rightarrow \infty} \sigma(s \cdot t) = \frac{1}{2} + sc$ for a constant $0 < c \leq \frac{1}{2}$ and is a.e. differentiable.*

Minimizer of the population loss. We first characterize the (unique) minimizers $\theta_L^* := \text{argmin}_{\theta} L(\theta, \sigma)$ of the population loss (9). We use the familiar ansatz that θ_L^* aligns with $u^* = \theta^* / \|\theta^*\|_2$, so $\theta = tu^*$. Using the formulation (9), we see that for $t^* := \|\theta^*\|_2$ and $\theta = tu^*$,

$$\begin{aligned} \nabla L(\theta, \sigma) &= -\mathbb{E}[\sigma(-\langle \theta, X \rangle)X\varphi_m(\langle X, \theta^* \rangle)] + \mathbb{E}[\sigma(\langle \theta, X \rangle)X(1 - \varphi_m(\langle X, \theta^* \rangle))] \\ &= -\mathbb{E}[\sigma(-tZ)X\varphi_m(t^*Z)] + \mathbb{E}[\sigma(tZ)X(1 - \varphi_m(t^*Z))] \end{aligned}$$

$$= (-\mathbb{E}[\sigma(-tZ)Z\varphi_m(t^*Z)] + \mathbb{E}[\sigma(tZ)Z(1 - \varphi_m(t^*Z))]) u^* = h_m(t)u^*, \quad (10)$$

where the final line uses the decomposition $X = Zu^* + W$ for the random vector $W \perp u^*$ independent of Z , and we recall expression (6) to define the *calibration function*

$$h_{t^*,m}(t) := \mathbb{E}[\sigma(tZ)Z(1 - \varphi_m(t^*Z))] - \mathbb{E}[\sigma(-tZ)Z\varphi_m(t^*Z)]. \quad (11)$$

The function h measures the “calibration” gap between the hypothesized link σ and the label probabilities φ_m . If we presume that a solution to $h_{t^*,m}(t) = 0$ exists, then tu^* minimizes $L(\theta, \sigma)$. In fact, such a solution exists and is unique (see Appendix D.1 for a proof):

Lemma 3.1. *Let Assumption A1 hold and $h = h_{t^*,m}$ be the gap function (11). Then there is a unique solution $t_m > 0$ to $h(t) = 0$, and $\theta_L^* = t_m u^*$ uniquely minimizes the loss (9). Define*

$$\begin{aligned} H_L(t) := & \mathbb{E}[(\sigma'(-tZ)\varphi_m(t^*Z) + \sigma'(tZ)(1 - \varphi_m(t^*Z)))Z^2]u^*u^{*\top} \\ & + \mathbb{E}[\sigma'(-tZ)\varphi_m(t^*Z) + \sigma'(tZ)(1 - \varphi_m(t^*Z))]P_{u^*}^\perp \Sigma P_{u^*}^\perp. \end{aligned} \quad (12)$$

Then the Hessian is $\nabla^2 L(\theta_L^*, \sigma) = H_L(t_m)$.

Asymptotic normality of the minimizers. To state asymptotic results when we have multiple labels, we define the average link function $\bar{\sigma}^* = \frac{1}{m} \sum_{j=1}^m \sigma_j^*$ and the three functions

$$\begin{aligned} \text{le}(Z) &:= \sigma(t_m Z)(1 - \bar{\sigma}^*(t^*Z)) - \sigma(-t_m Z)\bar{\sigma}^*(t^*Z), \\ \text{he}(Z) &:= \sigma'(-t_m Z)\varphi_m(t^*Z) + \sigma'(t_m Z)(1 - \varphi_m(t^*Z)), \\ v_j(Z) &:= \sigma_j^*(t^*Z)(1 - \sigma_j^*(t^*Z))(\sigma(t_m Z) + \sigma(-t_m Z))^2. \end{aligned} \quad (13)$$

The first, the link error le , measures mis-specification of the link σ relative to the average $\bar{\sigma}^*$.

The second, he , is a Hessian term, as $H_L(t_m) = \mathbb{E}[\text{he}(Z)Z^2]u^*u^{*\top} + \mathbb{E}[\text{he}(Z)]P_{u^*}^\perp \Sigma P_{u^*}^\perp$, and the

third measures variance for labeler j . We then prove the following theorem in Appendix D.

Theorem 1. Let Assumptions A1 and A3 hold, and let $\hat{\theta}_{n,m}$ be the multi-label estimator (8).

Define the shorthand $\bar{v} = \frac{1}{m} \sum_{j=1}^m v_j$. Then $\hat{\theta}_{n,m} \xrightarrow{a.s.} \theta_L^*$, and

$$\sqrt{n}(\hat{\theta}_n - \theta_L^*) \rightsquigarrow \mathbf{N} \left(0, \frac{\mathbb{E}[\text{le}(Z)^2 Z^2] + m^{-1} \mathbb{E}[\bar{v}(Z) Z^2]}{\mathbb{E}[\text{he}(Z) Z^2]^2} u^* u^{*\top} + \frac{\mathbb{E}[\text{le}(Z)^2] + m^{-1} \mathbb{E}[\bar{v}(Z)]}{\mathbb{E}[\text{he}(Z)]^2} (\mathbf{P}_{u^*}^\perp \Sigma \mathbf{P}_{u^*}^\perp)^\dagger \right).$$

Additionally, if $\hat{u}_n = \hat{\theta}_{n,m} / \|\hat{\theta}_{n,m}\|_2$ and t_m is the zero of the gap function $h_{t^*,m}(t) = 0$, then

$$\sqrt{n}(\hat{u}_n - u^*) \rightsquigarrow \mathbf{N} \left(0, \frac{1}{t_m^2} \frac{\mathbb{E}[\text{le}(Z)^2] + m^{-1} \mathbb{E}[\bar{v}(Z)]}{\mathbb{E}[\text{he}(Z)]^2} (\mathbf{P}_{u^*}^\perp \Sigma \mathbf{P}_{u^*}^\perp)^\dagger \right).$$

Theorem 1 exhibits two dependencies: the first on the link error terms $\mathbb{E}[\text{le}(Z)^2]$ —a bias term—and the second on the rescaled average variance $\frac{1}{m} \mathbb{E}[\bar{v}(Z)]$. The multi-label estimator recovers an optimal $O(1/m)$ covariance if the link errors are negligible, but if not, it necessarily has $O(1)$ asymptotic covariance. In the well-specified case that σ is symmetric and $\sigma = \bar{\sigma}^*$, the zero of the gap function (11) is evidently $t_m = t^* = \|\theta^*\|_2$, the error term $\text{le}(Z) = 0$, and $v_j(Z) = \sigma^*(t^* Z)(1 - \sigma^*(t^* Z))$, and by symmetry, $\sigma'(t) = \sigma'(-t)$ so that $\text{he}(Z) = \sigma'(t^* Z)$:

Corollary 3 (The well-specified case). Let the conditions above hold. Then

$$\sqrt{n}(\hat{u}_n - u^*) \rightsquigarrow \mathbf{N} \left(0, \frac{1}{m} \cdot \frac{1}{\|\theta^*\|_2^2} \frac{\mathbb{E}[\sigma(t^* Z)(1 - \sigma(t^* Z))]}{\mathbb{E}[\sigma'(t^* Z)]^2} \mathbf{P}_{u^*}^\perp \Sigma \mathbf{P}_{u^*}^\perp \right).$$

Asymptotic normality with majority vote. When we use the majority vote estimators, the asymptotics differ: no averaging reduces variance as m increases, even in a “well-specified” case. The asymptotic variance typically decreases m grows, but at a slower rate roughly related to the fraction of data where there is “signal”, as we discuss briefly following Corollary 2.

Theorem 2. Let Assumptions A1 and A3 hold, and let $\hat{\theta}_n = \hat{\theta}_{n,m}^{\text{mv}}$ be the general majority vote estimator (8). Let t_m be the zero of the gap function (11), solving $h_{t^*,m}(t) = 0$. Then $\hat{\theta}_n \xrightarrow{a.s.} \theta_L^* = t_m u^*$, and $\hat{u}_n = \hat{\theta}_n / \|\hat{\theta}_n\|_2$ satisfies

$$\sqrt{n}(\hat{u}_n - u^*) \rightsquigarrow \mathbf{N} \left(0, \frac{1}{t_m^2} \frac{\mathbb{E}[\sigma(-t_m |Z|)^2 \rho_m(t^* Z) + \sigma(t_m |Z|)^2 (1 - \rho_m(t^* Z))]}{\mathbb{E}[\text{he}(Z)]^2} (\mathbf{P}_{u^*}^\perp \Sigma \mathbf{P}_{u^*}^\perp)^\dagger \right).$$

We defer the proof to Appendix E.

In most cases, we take a symmetric link function, so that $\sigma(t) = 1 - \sigma(-t)$, and thus $\sigma'(t) = \sigma'(-t)$ and $\text{he}(z) = \sigma'(t_m z) \geq 0$. This simplifies the denominator in Theorem 2 to $\mathbb{E}[\sigma'(t_m Z)]^2$. Written differently, we may define a (scalar) variance-characterizing function C_m implicitly as follows: let $t_m = t_m(t)$ be a zero of $h_{t,m}(s) = \mathbb{E}[\sigma(sZ)Z(1 - \varphi_m(tZ))] - \mathbb{E}[\sigma(-sZ)Z\varphi_m(tZ)]$ in s , $h_{t,m}(t_m(t)) = 0$ so that t_m is a function of the size t (recall the gap (11)), and define

$$C_m(t) := \frac{1}{t_m^2} \frac{\mathbb{E}[\sigma(-t_m|Z|)^2 \rho_m(tZ) + \sigma(t_m|Z|)^2 (1 - \rho_m(tZ))]}{\mathbb{E}[\sigma'(t_m Z)]^2} \quad (14)$$

where $t_m = t_m(t)$ above is implicitly defined. Then

$$\sqrt{n}(\hat{u}_n - u^\star) \rightsquigarrow \mathbf{N}\left(0, C_m(t^\star) (\mathbf{P}_{u^\star}^\perp \Sigma \mathbf{P}_{u^\star}^\perp)^\dagger\right).$$

Each of our main results, including those on well-specified models, follows by characterizing the behavior of $C_m(t)$ in the asymptotics as $m \rightarrow \infty$ and the solution norm $t_m = \|\theta_L^\star\|_2$ scales, which the calibration gap (11) determines. The key is that the scaling with m varies with the fidelity of the model, behavior of the links σ , and the noise exponent ($\mathbf{M}_\beta$); our coming consequences of the master Theorems 1 and 2 reify this scaling.

3.2 Robustness to model mis-specification

Having established general convergence results for the multi-label $\hat{\theta}_{n,m}(\sigma)$ and majority vote $\hat{\theta}_{n,m}^{\text{mv}}(\sigma)$ estimators, we explicate their performance when the link σ is incorrect.

Multi-label estimator. To make interpretable statements about the multi-label estimator $\hat{\theta}_{n,m}$ in (8), assume identical links $\sigma_j^\star \equiv \sigma^\star \in \mathcal{F}_{\text{link}}$. A corollary of Theorem 1 then follows:

Corollary 4. *Let Assumptions A1 and A2 hold for some $\beta \in (0, \infty)$, and $t^\star = \|\theta^\star\|_2$. Then the calibration gap function (11) has unique positive zero $t_{\sigma^\star}$ such that $h_{t^\star,m}(t_{\sigma^\star}) = 0$, and*

the multi-label estimator (8) satisfies

$$\widehat{\theta}_{n,m} \xrightarrow{p} t_{\sigma^*} u^*.$$

For the mean $\bar{v} = \frac{1}{m} \sum_{j=1}^m v_j$ of the variances (13), the estimate $\widehat{u}_{n,m} = \widehat{\theta}_{n,m} / \|\widehat{\theta}_{n,m}\|_2$ satisfies

$$\sqrt{n} (\widehat{u}_{n,m} - u^*) \rightsquigarrow \mathbf{N} \left(0, \frac{\mathbb{E}[\mathbf{le}(Z)^2] + m^{-1} \mathbb{E}[\bar{v}(Z)]}{t_{\sigma^*}^2 \mathbb{E}[\mathbf{he}(Z)]^2} (\mathbf{P}_{u^*} \Sigma \mathbf{P}_{u^*})^\dagger \right).$$

In this simplified case, the asymptotic covariance remains of constant order in m unless $\mathbb{E}[\mathbf{le}(Z)^2] = 0$. In contrast, the majority vote estimator exhibits more robustness; this is perhaps expected, as Corollary 2 shows covariance scaling as $1/\sqrt{m}$ in the logistic link case, which is *a fortiori* misspecified for majority vote labels.

Majority vote estimator. For the majority-vote estimator, we relax our assumptions and allow σ_j^* to differ, showing how the broad conclusions Corollary 4 suggests hold in some generality: majority vote estimators achieve slower convergence than well-specified (maximum likelihood) estimators using each label but exhibit more robustness. To characterize the large m behavior, we require the following regularity conditions on the average link $\bar{\sigma}_m^* = \frac{1}{m} \sum_{j=1}^m \sigma_j^*$.

Assumption A4. For the sequence of link functions $\{\sigma_j \mid j \in \mathbb{N}\} \subset \mathcal{F}_{\text{link}}$, let $\bar{\sigma}_m^* = \frac{1}{m} \sum_{j=1}^m \sigma_j^*$. There exists $\bar{\sigma}^{*'}(0) > 0$ such that

$$(i) \quad \lim_{m \rightarrow \infty} \sqrt{m} \left(\bar{\sigma}_m^* \left(\frac{t}{\sqrt{m}} \right) - \frac{1}{2} \right) = \bar{\sigma}^{*'}(0)t, \quad \text{for each } t \in \mathbb{R}; \quad (15a)$$

$$(ii) \quad \liminf_{m \rightarrow \infty} \inf_{t \neq 0} \frac{|\bar{\sigma}_m^*(t) - \frac{1}{2}|}{\min\{|t|, 1\}} > 0; \quad (15b)$$

$$(iii) \quad \limsup_{t \rightarrow 0} \sup_{j \in \mathbb{N}} \left| \sigma_j^*(t) - \frac{1}{2} \right| = 0. \quad (15c)$$

These assumptions simplify if the links are identical: if $\sigma_j^* \equiv \sigma^*$, we only require σ^* is differentiable around 0 with $\sigma^{*'}(0) > 0$ and $|\sigma^*(t) - \frac{1}{2}| \gtrsim \min\{|t|, 1\}$.

We can apply Theorem 2 to obtain asymptotic normality for the estimator (8). Recall

$$\rho_m(t) := \mathbb{P}(Y^+ = \text{sign}(\langle X, \theta^* \rangle) \mid \langle X, \theta^* \rangle = t), \quad (16)$$

the probability of a correct majority vote given the margin $\langle X, \theta^* \rangle = t$, and the calibration gap function (11), which by a calculation resolves to the more convenient form

$$h(t) = h_{t^*, m}(t) = \mathbb{E}[\sigma(t|Z)|Z|(1 - \rho_m(t^*Z))] - \mathbb{E}[\sigma(-t|Z)|Z|\rho_m(t^*Z)].$$

The main challenge is to characterize the large m behavior for the asymptotic covariance function $C_m(t)$ the quantity (14) implicitly defines. We postpone details to Appendix F.3 and state the result, which follows from Theorem 2 and an asymptotic expansion of $C_m(t)$.

Proposition 3. *Let Assumptions A1 and A2 hold for some $\beta \in (0, \infty)$ with $\int_0^\infty z^{\beta-1} \sigma'(z) dz < \infty$ and $t^* = \|\theta^*\|_2$, and in addition that Assumption A4 holds and σ is symmetric. Then there are constants $a, b > 0$, depending only on β , c_Z , σ , and $\bar{\sigma}^{*'}(0)$, such that there is a unique $t_m \geq t_1 = t^*$ solving $h(t_m) = 0$, and for this t_m we have both*

$$\widehat{\theta}_{n,m}^{\text{mv}} \xrightarrow{p} t_m u^* \quad \text{and} \quad \lim_{m \rightarrow \infty} \frac{t_m}{t^* \sqrt{m}} = a.$$

Moreover, the covariance (14) has the form $C_m(t) = \frac{b}{(t\sqrt{m})^{2-\beta}}(1 + o_m(1))$, and

$$\sqrt{n} (\widehat{u}_{n,m}^{\text{mv}} - u^*) \rightsquigarrow \mathbf{N} \left(0, C_m(t^*) (\mathbf{P}_{u^*} \Sigma \mathbf{P}_{u^*})^\dagger \right).$$

Proposition 3 highlights the majority vote estimator's robustness: even when the link σ is more or less arbitrarily incorrect, the asymptotic covariance scales reasonably. The noise parameter β in Assumption A2, analogous to the noise exponent $(\mathbf{M}_\beta)$, plays an important role. In typical cases with $\beta = 1$ (e.g., when $X \sim \mathbf{N}(0, I_d)$), we see $C_m(t) \asymp \frac{1}{t\sqrt{m}}$. In noisier cases, corresponding to $\beta \downarrow 0$, majority vote provides benefits approaching a well-specified model; conversely, in “easy” cases where $\beta > 2$, majority vote estimators become *more* unstable, as they make the data nearly separable, making margin-type estimators volatile [6].

4 Fundamental estimation limits of majority vote labels

We complement our convergence guarantees via lower bounds for labelers with identical logistic links $\sigma_1^* = \dots = \sigma_m^* = \sigma^{\text{lr}}$, providing precise efficiency limits for estimation from majority labels Y^+ . These results highlight the robustness of aggregation-based estimators: the bounds match the rates $r(m) \asymp 1/\sqrt{m}$ in Table 1 even when the link σ is incorrect. As most of our results repose on classical asymptotics, we use local minimax theory to provide lower bounds, recalling a typical result combining [19, Lemma 6.6.5] and [30, Theorem 3.11.5]:

Corollary 5. *Let $\{\mathbb{P}_\theta\}$ be distributions with continuous Fisher Information $\mathsf{I}(\theta)$ in a neighborhood of θ^* and $W \sim \mathcal{N}(0, \mathsf{I}(\theta^*)^{-1})$. Then there exist probability densities $\pi_{c,n}$ supported on $\{\theta \in \mathbb{R}^d \mid \|\theta^* - \theta\|_2 \leq c/\sqrt{n}\}$ such that for any quasi-convex, symmetric, and bounded loss φ ,*

$$\liminf_{c \rightarrow \infty} \liminf_n \inf_{\hat{\theta}_n} \int \mathbb{E}_\theta[\varphi(\sqrt{n}(\hat{\theta}_n - \theta))] \pi_{c,n}(\theta) d\theta \geq \mathbb{E}[\varphi(W)].$$

If $T : \mathbb{R}^d \rightarrow \mathbb{R}^k$ is differentiable at θ^ with Jacobian $\dot{T}(\theta^*) \in \mathbb{R}^{k \times d}$, then additionally*

$$\liminf_{c \rightarrow \infty} \liminf_n \inf_{\hat{T}_n} \int \mathbb{E}_\theta[\varphi(\sqrt{n}(\hat{T}_n - T(\theta)))] \pi_{c,n}(\theta) d\theta \geq \mathbb{E}[\varphi(\dot{T}(\theta^*)W)].$$

The infima range over estimators $\hat{\theta}_n$ and $\hat{T}_n$ based on n i.i.d. observations from $\mathbb{P}_\theta$.

This corollary makes clear that (except at a Lebesgue measure-zero set of parameters θ) any estimator $\hat{\theta}_n$ with asymptotic variance Σ_θ satisfies $\Sigma_\theta \succeq \mathsf{I}(\theta)^{-1}$. We can therefore complete our theoretical analysis of fundamental limits by giving the Fisher Information matrix for binary classification models with majority vote data (see Appendix G for proofs).

As usual, letting Assumption A1 hold, we write $X = Zu^* + W$ with $W \perp u^*$. Let $t^* = \|\theta^*\|_2$. In this case, we may explicitly decompose the Fisher information for $\theta \in \mathbb{R}^d$.

Lemma 4.1. *The Fisher information $\mathsf{I}_m^{\text{mv}}(\theta)$ for the majority vote model $\{\mathbb{P}_{(X,Y^+)}^{\sigma^{\text{lr}}, \theta, m}\}$ satisfies*

$$\mathsf{I}_m^{\text{mv}}(\theta^*) = \mathbb{E} \left[\frac{\rho'_m(t^*|Z|)^2 Z^2}{\rho_m(t^*|Z|)(1 - \rho_m(t^*|Z|))} \right] \cdot u^* u^{*\top} + \mathbb{E} \left[\frac{\rho'_m(t^*|Z|)^2}{\rho_m(t^*|Z|)(1 - \rho_m(t^*|Z|))} \right] \cdot P_{u^*}^\perp \Sigma P_{u^*}^\perp.$$

By characterizing the large m behavior of the Fisher information, we can then apply Corollary 5 to understand the optimal efficiency of any estimator given majority vote data.

Theorem 3. *Let Assumption A1 hold, Assumption A2 hold with constants c_Z and β , and $t^\star = \|\theta^\star\|_2$. Then there are constants $a = a(c_Z, \beta), b = b(c_Z, \beta) > 0$ and functions*

$$A_m(t) = \frac{a}{t^2} \left(\frac{1}{t\sqrt{m}} \right)^\beta (1 + o_m(1)) \quad \text{and} \quad B_m(t) = \frac{b}{t^2} \left(\frac{1}{t\sqrt{m}} \right)^{\beta-2} (1 + o_m(1))$$

for which

$$\mathbf{I}_m^{\text{mv}}(\theta^\star) = A_m(t^\star) \cdot u^\star u^{\star\top} + B_m(t^\star) \cdot \mathbf{P}_{u^\star}^\perp \Sigma \mathbf{P}_{u^\star}^\perp.$$

Combining Theorem 3 with the local asymptotic minimax bounds, we obtain optimality for estimators of both $\theta^\star$ and the direction $u^\star$ as m gets large. The normalized quantity $\phi(\theta) := \theta / \|\theta\|_2$ satisfies $\dot{\phi}(\theta) = \|\theta\|_2^{-1} (I_d - \phi(\theta)\phi(\theta)^\top)$, yielding efficient limiting covariances

$$\begin{aligned} \Sigma_{\theta^\star} &:= \mathbf{I}_m^{\text{mv}}(\theta^\star)^{-1} \asymp t^{\star\beta+2} m^{\frac{\beta}{2}} \cdot u^\star u^{\star\top} + t^{\star\beta} m^{\frac{\beta-2}{2}} \cdot (\mathbf{P}_{u^\star}^\perp \Sigma \mathbf{P}_{u^\star}^\perp)^\dagger, \\ \Sigma_{u^\star} &:= \dot{\phi}(\theta^\star) \mathbf{I}_m^{\text{mv}}(\theta^\star)^{-1} \dot{\phi}(\theta^\star) \asymp t^{\star\beta-2} m^{\frac{\beta-2}{2}} \cdot (\mathbf{P}_{u^\star}^\perp \Sigma \mathbf{P}_{u^\star}^\perp)^\dagger, \end{aligned}$$

where we recall the shorthand $t^\star = \|\theta^\star\|_2$. (We use that for symmetric A, B with $AB = BA = 0$, $(A + B)^{-1} = A^\dagger + B^\dagger$; see Lemma C.1 in Appendix C.) Then combining Theorem 3 with Corollary 5 yields the following efficiency lower bound.

Corollary 6. *Let $\hat{\theta}_n$ and $\hat{u}_n$ be arbitrary estimators of $\theta^\star$ and $u^\star = \phi(\theta^\star)$ using the aggregated data $\{(X_i, Y_i^+)\}_{i=1}^n$. For any bounded and symmetric quasi-convex loss $\varphi : \mathbb{R}^d \rightarrow \mathbb{R}$,*

$$\lim_{c \rightarrow \infty} \liminf_n \sup_{\|\theta - \theta^\star\|_2 \leq c/\sqrt{n}} \mathbb{E}_\theta[\varphi(\sqrt{n}(\hat{u}_n - \phi(\theta)))] \geq \mathbb{E}[\varphi(W)], \quad \text{where } W \sim \mathbf{N}(0, \Sigma_{u^\star}).$$

If there exist T_θ or U_θ such that $\sqrt{n}(\hat{\theta}_n - \theta^\star) \rightsquigarrow T_{\theta^\star}$ (respectively, $\sqrt{n}(\hat{u}_n - u^\star) \rightsquigarrow U_{\theta^\star}$), then $\text{Cov}(T_\theta) \succeq \Sigma_{\theta^\star}$ and $\text{Cov}(U_\theta) \succeq \Sigma_{u^\star}$ for Lebesgue almost every $\theta^\star \in \mathbb{R}^d$.

This result carries several implications. First, the scaling in both $t^\star = \|\theta^\star\|_2$ and m for the asymptotic (achieved) covariance in Proposition 2 is optimal: we have $C_m(t) \asymp t^{\beta-2} m^{\frac{\beta-2}{2}}$ for the majority vote estimator $\widehat{\theta}_{n,m}^{\text{mv}}$. This optimal scaling holds even though the model may be mis-specified: estimators using the aggregated majority vote labels Y^+ are rate-optimal in the scale $\|\theta^\star\|_2$ and number of labelers m (and sample size n) no matter the margin-based loss. (Achieving the optimal constant requires a well-specified loss.) This robustness of estimators using aggregated data may motivate data-cleaning measures in dataset construction when it is difficult to model the particulars of the data generation process.

For any noise scale $\beta > 0$ on the margin variable $Z = \langle X, u^\star \rangle$ in Assumption A2, the inverse information $\Sigma_{\theta^\star} \rightarrow \infty$ as $m \rightarrow \infty$, so calibration becomes fundamentally harder for estimators using only aggregated labels even with known logistic link σ^{lr} . For example, in the Gaussian case ($\beta = 1$), then $\Sigma_{\theta^\star} = \Omega(m^{\frac{1}{2}})$ and $\Sigma_{u^\star} = O(m^{-\frac{1}{2}})$, matching Corollary 2’s scaling. The same phenomenon holds in the “low noise” regime where $\beta > 2$, so the margin variable Z is typically far from 0 (recall the exponent $(\mathbf{M}_\beta)$). In this case, $\Sigma_{u^\star} \rightarrow \infty$ as $m \rightarrow \infty$; estimators necessarily become non-robust when a paucity of data identifies the direction $u^\star$.

5 Experiments

We conclude the paper with several experiments to test the merit of the (sometimes implicit) methodological suggestions. We detail a few expected behaviors our theory suggests; if we fail to see them, then the model we have proposed is too unrealistic to inform practice. The results in Section 2 suggest the classification error should be lower for the non-aggregated algorithm $\widehat{\theta}_{n,m}^{\text{lr}}$, and the gap between algorithms (2) and (3) should become larger for less noisy problems. We only expect $\widehat{\theta}_{n,m}^{\text{lr}}$ to be calibrated, and the majority-vote (3) estimator’s calibration should

worsen as the number of labels m increases. So long as we model uncertainty with enough fidelity, Corollary 4 suggests that multi-label estimators should exhibit better performance than those using majority vote labels Y^+ .

To that end, we provide experiments on two real and one semi-synthetic dataset: the BlueBirds dataset [32] (Sec. 5.1), CIFAR-10H [23] (Appendix A), and a modified version of the CIFAR-10 dataset [18] (Sec. 5.2). We consider our two main algorithmic models:

- (i) The maximum likelihood estimator $\hat{\theta}_{n,m}^{\text{lr}}$ based on non-aggregated data in Eq. (2).
- (ii) The majority-vote based estimator $\hat{\theta}_{n,m}^{\text{mv}}$ of Eq. (3), our proxy for modern data pipelines.

A paucity of large-scale multi-label datasets that we know of precludes experiments on fully real datasets; the ImageNet creators [10, 27] no longer have intermediate label information. To simulate a larger scale data collection process, we create a semisynthetic dataset (Section 5.2) using the raw CIFAR-10 data and trained neural networks as labelers, giving us a semi-synthetic dataset with multiple labels. As the ground truth model is well-specified and we know the link function, it is also of practical interest to see how the aggregated labels from other crowdsourcing methods compare to majority vote and to knowing all individual labels.

5.1 BlueBirds

We begin with the BlueBirds dataset [32], a small dataset of 108 images with ResNet features. The task is to classify each image as one of *Indigo Bunting* or *Blue Grosbeak* (two similar-looking blue bird species). For each image, we have 39 labels, obtained through Amazon Mechanical Turk workers. We use an ImageNet pretrained ResNet50 model to generate image features, applying PCA to reduce the dimensionality from $d_{\text{init}} = 2048$ to $d = 25$.

We repeat the following experiment $T = 100$ times. For each number $m = 1, \dots, 35$ of

labelers, we fit the multi-label logistic model (2) and the majority vote estimator (3), finding calibration and classification errors using 10-fold cross validation. We measure calibration error on an example x by $|\text{logit}(\tilde{p}(x)) - \text{logit}(\hat{p}(x))|$, where $\tilde{p}(x)$ is the predicted probability, $\hat{p}(x)$ the empirical labeler probability, and $\text{logit}(p) = \log \frac{p}{1-p}$; we measure classification error on example x with labels $(y_1, \dots, y_m)$ by $\frac{1}{m} \sum_{j=1}^m \mathbb{1}\{y_j \neq \text{sign}(\tilde{p}(x) - \frac{1}{2})\}$, giving an inherent noise floor because of labeler uncertainty. We report the results in Figure 1. These plots corroborate our theoretical predictions: as the number of labelers m increases, both the majority vote method and the full label method exhibit improved classification error, but considering all labels gives a (significant) improvement in accuracy and in calibration error.

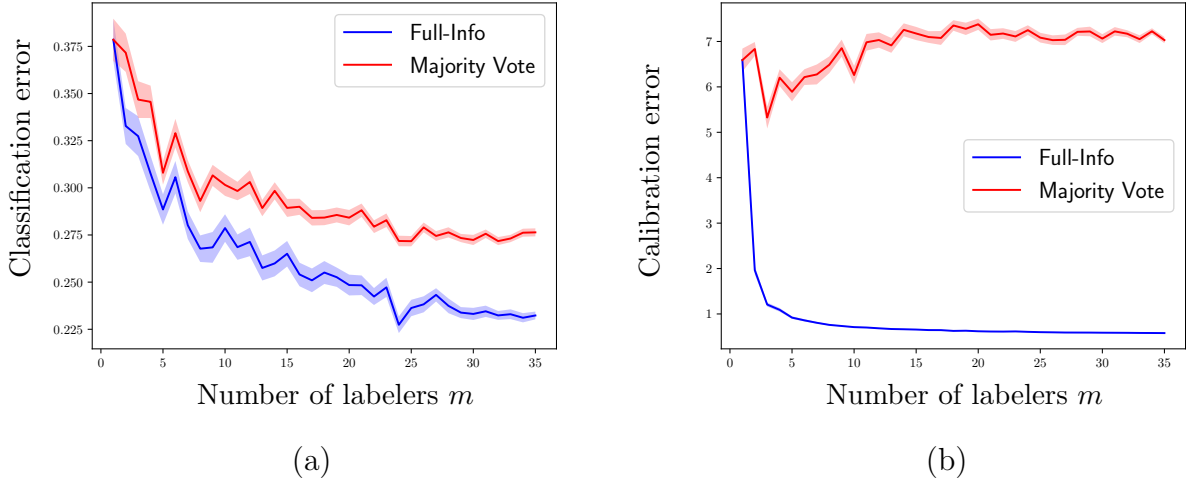


Figure 1. Experiments on BlueBirds dataset. (a) Classification error. (b) Calibration error $|\text{logit}(\tilde{p}) - \text{logit}(p)|$ with ResNet features reduced via PCA to dimension $d = 25$. Error bars show 2 standard error confidence bands over $T = 100$ trials.

5.2 Crowdsourcing methods on semisynthetic CIFAR-10 labels

We adapt the original CIFAR-10 dataset [18], which consists of 6000 32×32 images from each of $k = 10$ classes (60,000 total images). To mimic collecting messy data, rather than

the single-label in the base CIFAR-10 data, we construct pseudo-labelers using a pretrained eighteen layer residual network $f : \mathcal{X} \rightarrow \mathbb{R}^k$ (ResNet18 [16]) whose accuracy we can adjust. For $s \in \mathbb{R}^k$, define $\text{softmax}(s) = [e^{s_y} / \sum_{l=1}^k e^{s_l}]_{y=1}^k$. Then for an input $x \in \mathcal{X}$, the model outputs a score $f(x) = \Theta^{\star\top} \phi(x) \in \mathbb{R}^k$, where $\Theta^{\star} = [\theta_1^{\star} \cdots \theta_k^{\star}] \in \mathbb{R}^{d \times k}$ is a matrix of per-class weights and $\phi : \mathcal{X} \rightarrow \mathbb{R}^d$ represents the second-to-last layer outputs of the neural network. As we fit linear models, we take the outputs $\phi(x_i)$ as data, so $X_i = \phi(x_i)$ are the covariates for the i th image x_i , and $\Theta^{\star}$ is the ground-truth parameter. We construct labels $Y_{ij} \in \{1, \dots, k\}$, $j = 1, \dots, m$ by choosing the constant $\alpha > 0$ to model labeler expertise, drawing

$$\mathbb{P}(Y_{ij} = y \mid X_i = x) = \text{softmax}(\alpha f(x))_y = \frac{\exp(\alpha \langle \theta_y^{\star}, \phi(x) \rangle)}{\sum_{l=1}^k \exp(\alpha \langle \theta_l^{\star}, \phi(x) \rangle)}.$$

We vary α in experiments to adjust that the median value of $p_i := \max_{y \in [m]} \text{softmax}(\alpha f(x_i))_y$, where $p_i \in (1/k, 1)$. In the “expert labelers” case, we take $\alpha \rightarrow \infty$ so that $\text{median}(\{p_i\}) \rightarrow 1$, while the “crowd labelers” case corresponds to $\alpha \downarrow 0$ and $\text{median}(\{p_i\}) \rightarrow 1/k$.

We fit $\hat{\theta}_{n,m}^{\text{mv}}$ and $\hat{\theta}_{n,m}^{\text{mle}}$ using the multiclass logistic loss. To test our predictions, we also investigate the standard Dawid-Skene (DS) [9] and GLAD [33] crowdsourcing methods. The DS method models the probability that labeler j assigns label y' given correct label y through a matrix $\alpha_j \in \mathbb{R}^{k \times k}$ with $p_j(y' \mid y) = \frac{1}{1 + e^{-\alpha_j(y', y)}}$, while GLAD estimates the probability that labeler j correctly identifies the label of an example i via $p_{ij} = \frac{1}{1 + e^{-\beta_i \alpha_j}}$, where $\alpha_j \in [-\infty, \infty]$ indicates labeler j ’s ability and $\beta_i \in [0, \infty]$ ambiguity in example i . Each method uses EM to approximately maximize the likelihood $\prod_{i=1}^n \prod_{j=1}^m \mathbb{P}(Y_{ij} \mid \alpha_j, \beta_i)$ (letting $\beta_i = 1$ for DS) of the observed labels Y_{ij} , initializing α to maximize the likelihood $\prod_{i=1}^n \prod_{j=1}^m \mathbb{P}(Y_{ij} \mid Y_i^+, \alpha_j, \beta_i)$ assuming the majority votes Y_i^+ are correct and $\beta_i = 1$. Given estimates $\hat{\alpha}$ and $\hat{\beta}$, one may impute soft labels $\hat{p}_i \in \mathbb{R}_k^+$ for each example i , where $\hat{p}_{iy}$ indicates the method’s estimated probability that example i is of class y , and hard labels $\hat{Y}_i = \arg\max_y \hat{p}_{iy}$. The labels yield

estimators from, respectively, hard and soft labels via

$$\hat{\theta}_h = \operatorname{argmin}_{\theta_1, \dots, \theta_k} \sum_{i=1}^n \log \left(\sum_{l=1}^k e^{\langle X_i, \theta_l - \theta_{\hat{Y}_i} \rangle} \right) \quad \text{and} \quad \hat{\theta}_s = \operatorname{argmin}_{\theta_1, \dots, \theta_k} \sum_{i=1}^n \sum_{y=1}^k \hat{p}_{iy} \log \left(\sum_{l=1}^k e^{\langle X_i, \theta_l - \theta_y \rangle} \right).$$

We let $\hat{\theta}_{n,m}^{\text{DS}}$ and $\hat{\theta}_{n,m}^{\text{GLAD}}$ denote the corresponding estimators using hard labels, and $\hat{\theta}_{n,m}^{\text{DS-P}}$ and $\hat{\theta}_{n,m}^{\text{GLAD-P}}$ those trained with the estimated probabilities $\hat{p}_i$. Our experiments use Crowd-Kit [28].

We report the results in Fig. 2. As we see, when the model is well-specified, the MLE $\hat{\theta}_{n,m}^{\text{mle}}$ outperforms aggregation methods and black-box crowdsourcing models that generate soft labels. Crowdsourced $\hat{\theta}_{n,m}^{\text{DS}}$ and $\hat{\theta}_{n,m}^{\text{GLAD}}$ yield smaller test error than $\hat{\theta}_{n,m}^{\text{mv}}$, and using the soft labels, $\hat{\theta}_{n,m}^{\text{DS-P}}$ and $\hat{\theta}_{n,m}^{\text{GLAD-P}}$ further reduces the error, suggesting benefits to using soft labels from crowdsourcing for training even if the underlying model is unknown.

6 Discussion and extensions

Semi-parametric approaches. Our analysis separates likelihood-based approaches using all the labels, as in (2), and majority vote estimators, which are robust but have slower convergence, as in Proposition 3. While we mainly focus on efficiency bounds for procedures we view as closer to practice, in this section, we present a few extensions to semiparametric approaches, more as an amuse-bouche to motivate further efficiency investigations rather than as a final word. In distinction from standard results in semiparametric theory (e.g. [29, Ch. 25] or [4]), the results require little more than consistent estimation of the links σ_j^* to recover (rate optimal) $1/m$ scaling in the asymptotic covariance.

Let $\vec{\sigma} = (\sigma_1, \dots, \sigma_m)$ be potentially distinct labeler link functions (with σ_j^* denoting the true links). Then for $(x, y) \in \mathbb{R}^d \times \{\pm 1\}^m$ define the link-based loss

$$\ell_{\vec{\sigma}, \theta}(y \mid x) := \frac{1}{m} \sum_{j=1}^m \ell_{\sigma_j, \theta}(y_j \mid x).$$

To eliminate ambiguity in $\vec{\sigma}$, we normalize $\|\theta^*\|_2 = 1$, and then define the population loss

$$L(\theta, \vec{\sigma}) := \frac{1}{m} \sum_{j=1}^m \mathbb{E}[\ell_{\sigma_j, \theta}(Y_j | X)].$$

For the sequence $\{\vec{\sigma}_n\} \subset \mathcal{F}_{\text{link}}^{\text{sp}}$ of (estimated) links and data (X_i, Y_i) , $Y_i = (Y_{i1}, \dots, Y_{im})$, we define the semi-parametric estimator

$$\hat{\theta}_{n,m}^{\text{sp}} := \underset{\theta \in \mathbb{R}^d}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^n \ell_{\vec{\sigma}_n, \theta}(Y_i | X_i).$$

Both consistency and asymptotic normality hold under appropriate convergence of the link functions (see Appendix H for details). Define the distance $d_{\mathcal{F}_{\text{link}}}$ on $\mathbb{R}^d \times \mathcal{F}_{\text{link}}$ by

$$d_{\mathcal{F}_{\text{link}}}((\theta_1, \vec{\sigma}_1), (\theta_2, \vec{\sigma}_2)) := \|\theta_1 - \theta_2\|_2 + \|\vec{\sigma}_1(-Y\langle X, u^* \rangle) - \vec{\sigma}_2(-Y\langle X, u^* \rangle)\|_{L^2(\mathbb{P})}. \quad (17)$$

Then the next master result for semi-parametric approaches characterizes the asymptotic behavior of the semi-parametric estimator $\hat{\theta}_{n,m}^{\text{sp}}$ using the variance function

$$C_{m, \vec{\sigma}^*} := \frac{1}{m} \cdot \frac{\frac{1}{m} \sum_{j=1}^m \mathbb{E}[\sigma_j^*(Z)(1 - \sigma_j^*(Z))]}{(\frac{1}{m} \sum_{j=1}^m \mathbb{E}[\sigma_j^{*'}(Z)])^2}.$$

Theorem 4. *Let Assumption A1 hold and assume $|Z| > 0$ has nonzero and continuous density $p(z)$ on $(0, \infty)$. Suppose the sequence $\{\vec{\sigma}_n\}$ is consistent, i.e.,*

$$\|\vec{\sigma}_n(-Y\langle X, u^* \rangle) - \vec{\sigma}^*(-Y\langle X, u^* \rangle)\|_{L^2(\mathbb{P})} \xrightarrow{p} 0,$$

the mapping $(\theta, \vec{\sigma}) \mapsto \nabla_{\theta}^2 L(\theta, \vec{\sigma})$ is continuous for $d_{\mathcal{F}_{\text{link}}}$ at $(u^, \vec{\sigma}^*)$, and $\mathbb{E}[\|X\|_2^4] < \infty$. Then $\sqrt{n}(\hat{\theta}_{n,m}^{\text{sp}} - u^*)$ is asymptotically normal, and the estimator $\hat{u}_{n,m}^{\text{sp}} = \hat{\theta}_{n,m}^{\text{sp}} / \|\hat{\theta}_{n,m}^{\text{sp}}\|_2$ satisfies*

$$\sqrt{n}(\hat{u}_{n,m}^{\text{sp}} - u^*) \rightsquigarrow \mathbf{N}\left(0, C_{m, \vec{\sigma}^*} (\mathbf{P}_{u^*}^{\perp} \Sigma \mathbf{P}_{u^*}^{\perp})^{\dagger}\right).$$

Notably, Theorem 4 exhibits optimal $1/m$ covariance scaling whenever $\mathbb{E}[\sigma_j^{*'}(Z)] \gtrsim 1$.

We give two example applications of the general theory in Appendix H: the first analyzing a full pipeline for a single index model setting, which robustly estimates direction u^* , the

link σ^* , and then re-estimates θ^* ; the second assuming a stylized black-box crowdsourcing mechanism that provides estimates of labeler reliability, highlighting how even in some crowdsourcing scenarios, there could be substantial advantages to using full label information.

Future work. In spite of the technical detail we require to prove our results, we view this work as almost preliminary and hope that it inspires further work on the full pipeline of statistical machine learning, from dataset creation to model release. Many questions remain.

On the theoretical side, we focus on a stylized model of label aggregation, with majority vote—with the exception of the crowdsourcing model above—functioning as the stand-in for more sophisticated aggregation strategies. It seems challenging to show that *no* aggregation strategy can work as well as multi-label strategies; it would be interesting to delineate the benefits and drawbacks of more sophisticated denoising. Additionally, the insights make low-dimensional asymptotic predictions; investigating high-dimensional scaling might yield new perspectives as, for, classification datasets often become separable [6].

On the methodological and applied side, given the extent to which the test/challenge dataset methodology drives progress in machine learning [11], it seems that developing newer datasets to incorporate labeler uncertainty could yield substantial benefits. A particular refrain is that modern deep learning methods are overconfident in their predictions [13]; perhaps by calibrating them to *labeler* uncertainty we could improve their robustness and performance. Large-scale models now frequently use only distantly supervised labels [25], making the study of appropriate data cleaning and collection more timely. We look forward to deeper investigations of the foundations of the full practice of statistical machine learning.

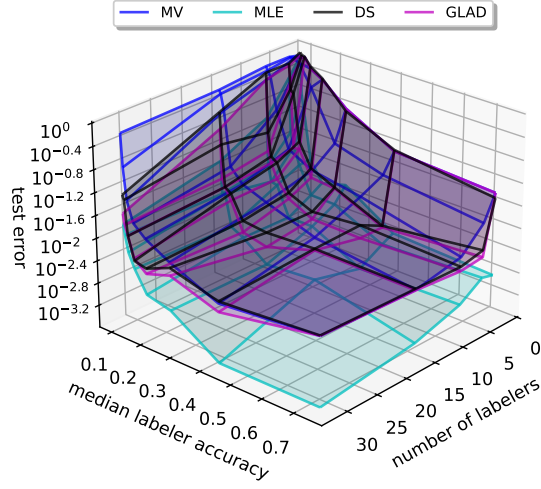
References

- [1] A. Asuncion and D. J. Newman. UCI machine learning repository, 2007. URL <http://www.ics.uci.edu/~mlearn/MLRepository.html>.
- [2] P. L. Bartlett, M. I. Jordan, and J. McAuliffe. Convexity, classification, and risk bounds. *Journal of the American Statistical Association*, 101:138–156, 2006.
- [3] A. Beygelzimer, P. Liang, J. Wortman Vaughan, and Y. Dauphin. NeurIPS 2021 datasets and benchmarks track.
- [4] P. Bickel, C. A. J. Klaassen, Y. Ritov, and J. Wellner. *Efficient and Adaptive Estimation for Semiparametric Models*. Springer Verlag, 1998.
- [5] S. Boucheron, O. Bousquet, and G. Lugosi. Theory of classification: a survey of some recent advances. *ESAIM: Probability and Statistics*, 9:323–375, 2005.
- [6] E. Candès and P. Sur. A modern maximum-likelihood theory for high-dimensional logistic regression. *Proceedings of the National Academy of Sciences*, 116(29):14516–14525, 2019.
- [7] M. Charikar, J. Steinhardt, and G. Valiant. Learning from untrusted data. In *Proceedings of the Forty-Ninth Annual ACM Symposium on the Theory of Computing*, 2017.
- [8] N. Codella, V. Rotemberg, P. Tschandl, M. E. Celebi, S. Dusza, D. Gutman, B. Helba, A. Kalloo, K. Liopyris, M. Marchetti, H. Kittler, and A. Halpern. Skin lesion analysis toward melanoma detection 2018: A challenge hosted by the International Skin Imaging Collaboration (ISIC). *arXiv:1902.03368 [cs.CV]*, 2019.
- [9] A. Dawid and A. Skene. Maximum likelihood estimation of observer error-rates using the EM algorithm. *Journal of the Royal Statistical Society*, 28:20–28, 1979.

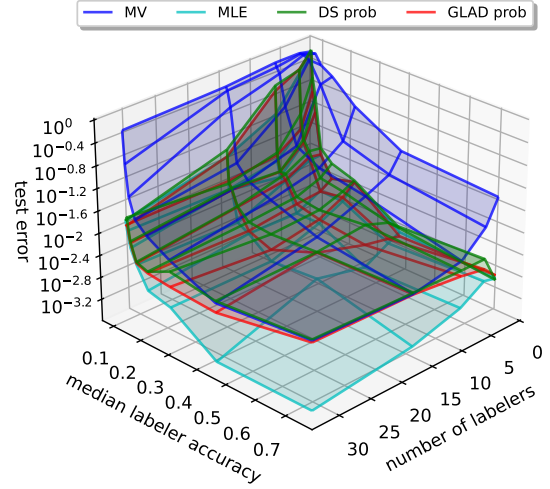
- [10] J. Deng, W. Dong, R. Socher, L. Li, K. Li, and L. Fei-Fei. ImageNet: a large-scale hierarchical image database. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 248–255, 2009.
- [11] D. L. Donoho. 50 years of data science. *Journal of Computational and Graphical Statistics*, 26(4):745–766, 2017.
- [12] S. Y. Gadre, G. Ilharco, A. Fang, J. Hayase, G. Smyrnis, T. Nguyen, R. Marten, M. Wortsman, D. Ghosh, J. Zhang, E. Orgad, R. Entezari, G. Daras, S. Pratt, V. Ramanujan, Y. Bitton, K. Marathe, S. Mussmann, R. Vencu, M. Cherti, R. Krishna, P. W. Koh, O. Saukh, A. Ratner, S. Song, H. Hajishirzi, A. Farhadi, R. Beaumont, S. Oh, A. Dimakis, J. Jitsev, Y. Carmon, V. Shankar, and L. Schmidt. DataComp: In search of the next generation of multimodal datasets. In *Advances in Neural Information Processing Systems 36*, 2023.
- [13] I. J. Goodfellow, J. Shlens, and C. Szegedy. Explaining and harnessing adversarial examples. In *Proceedings of the Third International Conference on Learning Representations*, 2015.
- [14] M. Hardt and B. Recht. *Patterns, Predictions, and Actions: A story about machine learning*. Princeton University Press, 2022. Available at <https://mlstory.org/>.
- [15] T. Hastie, R. Tibshirani, and J. Friedman. *The Elements of Statistical Learning*. Springer, second edition, 2009.
- [16] K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 770–778, 2016.

- [17] D. Karger, S. Oh, and D. Shah. Budget-optimal task allocation for reliable crowdsourcing systems. *Operations Research*, 62(1):1–24, 2014.
- [18] A. Krizhevsky and G. Hinton. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.
- [19] L. Le Cam and G. L. Yang. *Asymptotics in Statistics: Some Basic Concepts*. Springer, 2000.
- [20] D. Lewis, Y. Yang, T. Rose, and F. Li. RCV1: A new benchmark collection for text categorization research. *Journal of Machine Learning Research*, 5:361–397, 2004.
- [21] E. Mammen and A. B. Tsybakov. Smooth discrimination analysis. *Annals of Statistics*, 27:1808–1829, 1999.
- [22] M. P. Marcus, B. Santorini, and M. A. Marcinkiewicz. Building a large annotated corpus of English: the Penn Treebank. *Computational Linguistics*, 19:313–330, 1994.
- [23] J. C. Peterson, R. M. Battleday, T. L. Griffiths, and O. Russakovsky. Human uncertainty makes classification more robust. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 9617–9626, 2019.
- [24] E. A. Platanios, M. Al-Shedivat, E. Xing, and T. Mitchell. Learning from imperfect annotations. *arXiv:2004.03473 [cs.LG]*, 2020.
- [25] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning*, 2021.

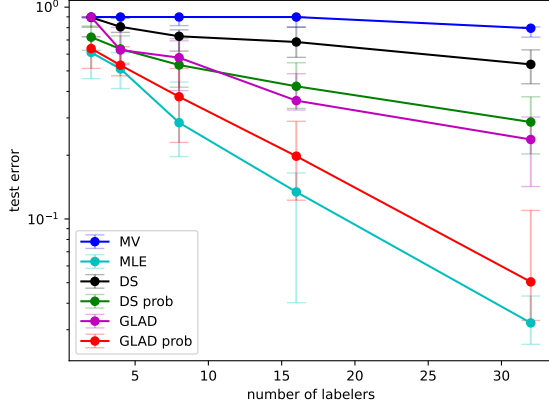
- [26] A. Ratner, S. H. Bach, H. Ehrenberg, J. Fries, S. Wu, and C. Ré. Snorkel: rapid training data creation with weak supervision. *Proceedings of the VLDB Endowment*, 11(3):269–282, 2017.
- [27] O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein, A. C. Berg, and L. Fei-Fei. ImageNet large scale visual recognition challenge. *International Journal of Computer Vision*, 115(3):211–252, 2015.
- [28] D. Ustalov, N. Pavlichenko, and B. Tseitlin. Learning from crowds with crowd-kit. *Journal of Open Source Software*, 9(96):6227, 2024.
- [29] A. W. van der Vaart. *Asymptotic Statistics*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, 1998.
- [30] A. W. van der Vaart and J. A. Wellner. *Weak Convergence and Empirical Processes: With Applications to Statistics*. Springer, New York, 1996.
- [31] V. Vapnik. *The Nature of Statistical Learning Theory*. Springer, 1995.
- [32] P. Welinder, S. Branson, P. Perona, and S. Belongie. The multidimensional wisdom of crowds. In *Advances in Neural Information Processing Systems 23*, pages 2424–2432, 2010.
- [33] J. Whitehill, T. Wu, J. Bergsma, J. Movellan, and P. Ruvolo. Whose vote should count more: Optimal integration of labels from labelers of unknown expertise. *Advances in Neural Information Processing Systems 22*, 22:2035–2043, 2009.



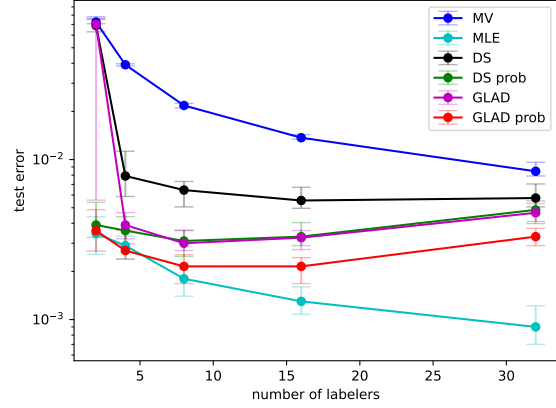
(a)



(b)



(c)



(d)

Figure 2. Experiments on semisynthetic CIFAR-10 dataset. Median labeler accuracy represents the median of $p_i = \max_y \text{softmax}(\alpha f(x_i))_y$ on the training data. Results are averaged over 20 trials, and vertical axes give mis-classification rate from (synthetic) ground-truth labels on held-out test set. Legend keys correspond to maximum likelihood $\hat{\theta}_{n,m}^{\text{mle}}$ (MLE), majority vote $\hat{\theta}_{n,m}^{\text{mv}}$ (MV), hard-labeled Dawid-Skene (DS) and GLAD crowdsourced estimators $\hat{\theta}_{n,m}^{\text{DS}}$ and $\hat{\theta}_{n,m}^{\text{GLAD}}$, and soft-labeled Dawid-Skene (DS prob) and GLAD (GLAD prob) estimators $\hat{\theta}_{n,m}^{\text{DS-P}}$ and $\hat{\theta}_{n,m}^{\text{GLAD-P}}$. (a) Comparison of methods using hard labels. (b) Comparison with crowdsourced estimated soft-labels. (c) Number of labelers m versus test error for median accuracy $p = .105$ (noisy labelers). (d) Number of labelers m versus test error for median accuracy $p = .4$. Both (c) and (d) report 95% error bars over the trials.